

# Socioeconomic Impact Evaluation of Terrestrial Protected Areas in Madagascar based on large national surveys

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**Abstract** Protected Areas are the most widely used tool for biodiversity conservation. However, their implementation raises concerns about the well-being of local populations, especially when they are very poor and dependent on natural resources, as is the case in Madagascar. We are using geolocated socioeconomic surveys spanning a period of protected area creation (2008-2021), and extending the analysis to the previous period (1997-2008) in order to verify the credibility of the comparison. Socio-environmental indicators derived from spatial data are used to match rural areas with similar probabilities of being affected by the creation of protected areas, before estimating the effects using an approach combining matching and difference-in-difference. The analysis, conducted in accordance with a pre-analysis plan submitted prior to any analysis,

shows that our empirical model would have detected average effects of at least 7.5 percentiles of the wealth index for rural households. The results indicate that, over the period considered, the creation of protected areas had no statistically significant average effect on the well-being of households located within 10 km of protected areas created after 2008, as measured by the intra-cluster dispersion of this index. Robustness test conducted using other survey sources support this conclusion. Further work is needed to determine whether this lack of effect reflects a trade-off between positive and negative impacts, high heterogeneity of effects, or limited implementation of conservation measures in a context of weak governance and underfunding of protected areas.

J'ai inclus les graphs et tables sous formes d'objets R dans `soumission_articles/figures` et `soumission_articles/tables` Voici les codes à copier-coller pour les intégrer dans ton document quarto.

Les options de rendu sont notés par des balises précédées de `#|` au début de chaque chunk R Il est aussi possible de gérer ces options globalement en les mettant dans le cartouche yaml au début du document quarto, mais là on va faire simple.

Attention car les chemins d'accès sont différents selon d'où tu lances le render (en cliquant sur render dans RStudio ou en lançant dans le terminal : dans certains cas, il faut ajouter ou enlever `soumission_articles/manuscript` au début des chemins d'accès aux fichiers).

Nota bene sur la mise en forme dans quarto.

# 1 Results

## 1.1 Overall impact on livelihoods

### 1.1.a Impact in 2x2 DiD

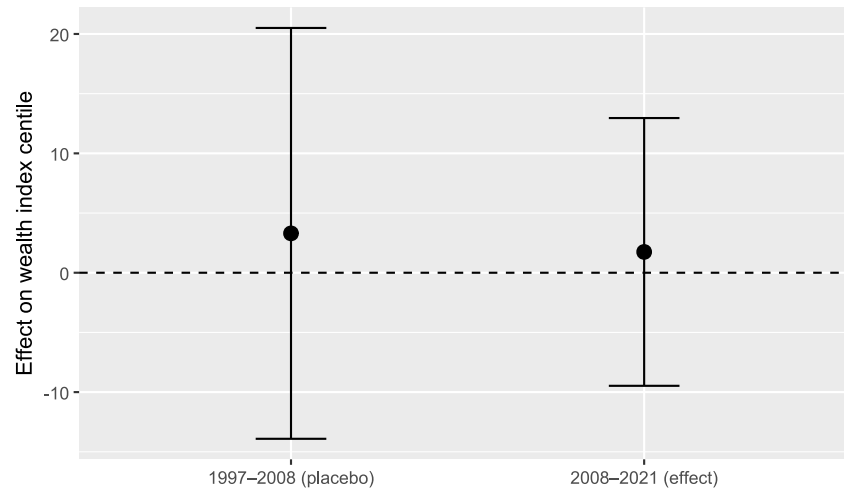


Figure 1: Impact on well-being: 2x2 DiD impact on local population (< 10km)

### 1.1.b Staggered DiD: static

Static effect

Table 1: Static treatment effect estimates on well-being

|              | did2s (static)                    |
|--------------|-----------------------------------|
| treat_on = 1 | 4.519                             |
|              | (4.234)                           |
| Num.Obs.     | 4246                              |
| R2           | 0.006                             |
| R2 Adj.      | 0.006                             |
| AIC          | 41025.8                           |
| BIC          | 41032.2                           |
| RMSE         | 30.32                             |
| Std.Errors   | Corrected Clustered (cluster_uid) |

### 1.1.c Staggered DiD: dynamic event study

(NB: see if this goes here or in the methods section) For the event study, each observation is indexed by its relative year to treatment, defined as the difference between the survey

year and the year in which the protected area was established for treated locations. This transformation allows outcomes to be compared along a common event time dimension across cohorts. Because surveys are conducted at irregular intervals and some relative years are represented by very few observations, we group relative years into two-year bins. For example, relative years  $-2$  and  $-1$  are grouped together, as are  $+4$  and  $+5$ . Observations occurring more than ten years before or after treatment are further aggregated into open-ended bins, as shown in Table 2. This binning strategy increases the number of observations contributing to each event-time coefficient, reduces noise at the extremes, and improves the stability and interpretability of the estimated dynamic effects.

**A inclure dans l'annexe**

Table 2: Binning of relative years-to-treatment for the event study

| Rel. year | Group  | Clusters | Obs.  | Bin       | Clusters | Obs.  |
|-----------|--------|----------|-------|-----------|----------|-------|
| -18       | Treat. | 7        | 218   | min to -9 | 18       | 552   |
| -16       | Treat. | 2        | 55    |           |          |       |
| -13       | Treat. | 1        | 27    |           |          |       |
| -12       | Treat. | 1        | 30    |           |          |       |
| -11       | Treat. | 4        | 129   |           |          |       |
| -10       | Treat. | 1        | 31    |           |          |       |
| -9        | Treat. | 2        | 62    |           |          |       |
| -7        | Treat. | 23       | 696   | -8 to -7  | 23       | 696   |
| -6        | Treat. | 1        | 32    | -6 to -5  | 3        | 93    |
| -5        | Treat. | 2        | 61    |           |          |       |
| -4        | Treat. | 12       | 349   | -4 to -3  | 14       | 413   |
| -3        | Treat. | 2        | 64    |           |          |       |
| -2        | Treat. | 26       | 817   | -2 to -1  | 28       | 878   |
| -1        | Treat. | 2        | 61    |           |          |       |
| 0         | Treat. | 12       | 368   | 0 to 1    | 27       | 846   |
| 1         | Treat. | 15       | 478   |           |          |       |
| 2         | Treat. | 1        | 32    | 2 to 3    | 21       | 652   |
| 3         | Treat. | 20       | 620   |           |          |       |
| 4         | Treat. | 3        | 89    | 4 to 5    | 13       | 411   |
| 5         | Treat. | 10       | 322   |           |          |       |
| 6         | Treat. | 37       | 1,141 | 6 to 7    | 38       | 1,172 |
| 7         | Treat. | 1        | 31    |           |          |       |
| 8         | Treat. | 14       | 449   | 8 to 9    | 16       | 500   |
| 9         | Treat. | 2        | 51    |           |          |       |
| 10        | Treat. | 7        | 222   | 10 to max | 27       | 851   |
| 11        | Treat. | 8        | 260   |           |          |       |
| 13        | Treat. | 12       | 369   |           |          |       |
| NA        | Ctrl.  | 282      | 7,064 | NA        | 282      | 7,064 |

Event study results. You can refer to the figure by mentioning its name as Figure 2.

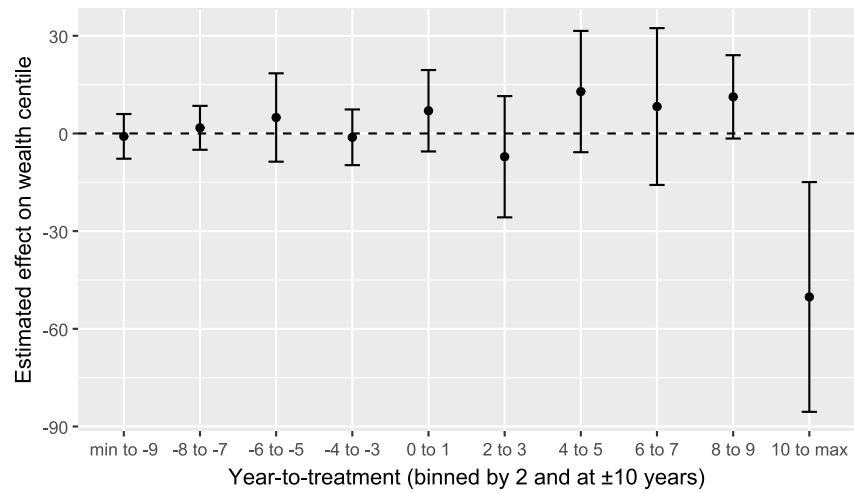


Figure 2: Impact on well-being: Event-study with 2-year bins

## 1.2 Effect on inequalities

### 1.2.a Impact in 2x2 DiD

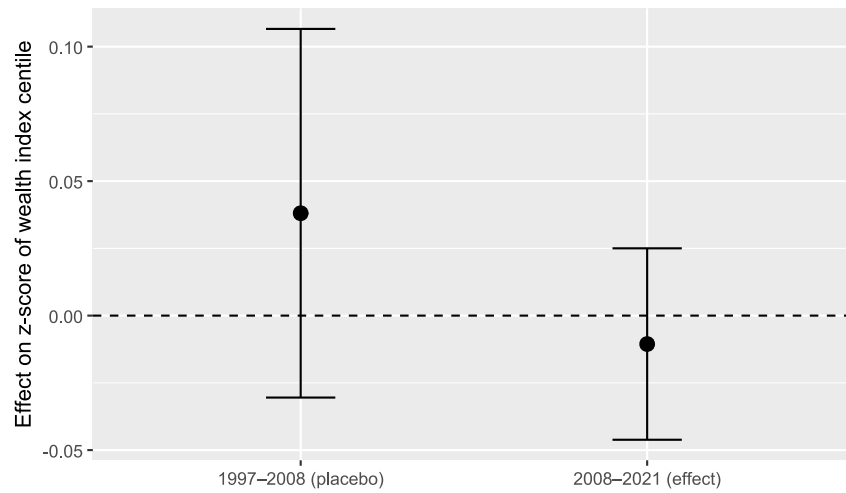


Figure 3: Impact on inequalities: 2x2 DiD impact on local population (< 10km)

### 1.2.b Staggered DiD: static

Static effect

Table 3: Static treatment effect estimates on inequalities

|              | did2s (static)                    |
|--------------|-----------------------------------|
| treat_on = 1 | -0.007                            |
|              | (0.019)                           |
| Num.Obs.     | 4246                              |
| R2           | -0.000                            |
| R2 Adj.      | -0.000                            |
| AIC          | 7251.5                            |
| BIC          | 7257.9                            |
| RMSE         | 0.57                              |
| Std.Errors   | Corrected Clustered (cluster_uid) |

### 1.2.c Staggered DiD: dynamic event study

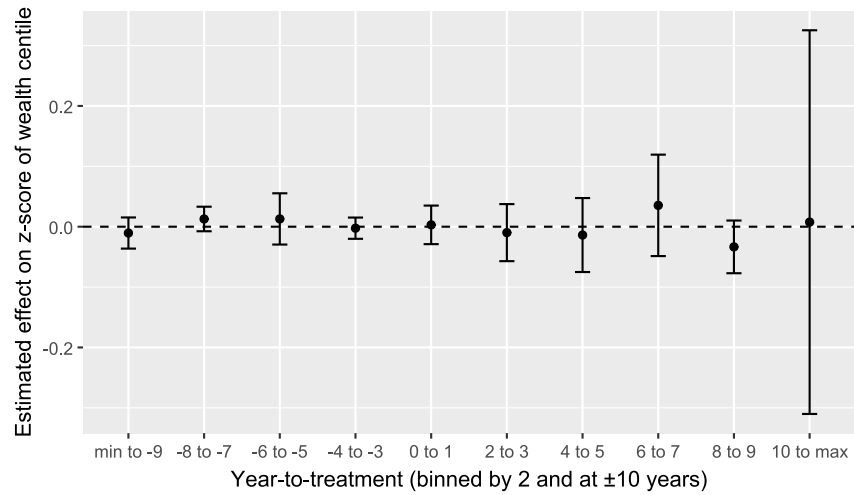


Figure 4: Impact on inequalities: Event-study with 2-year bins

## 1.3 Heterogeneity

### 1.3.a Impact in 2x2 DiD

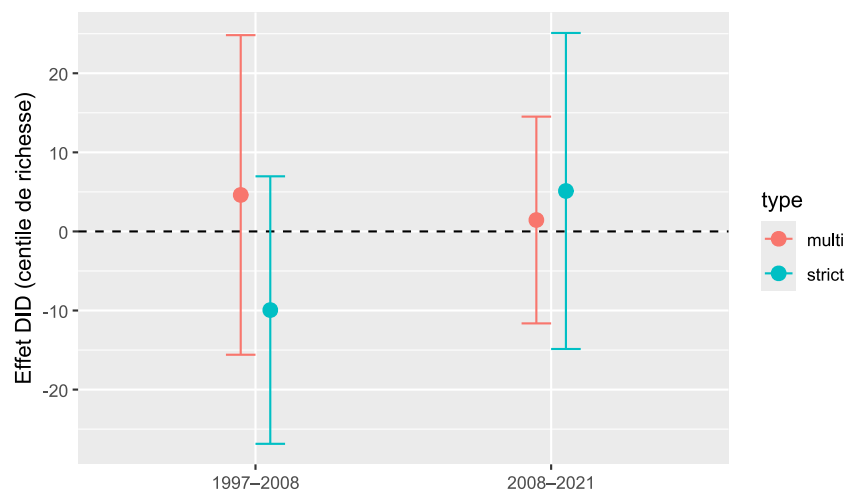


Figure 5: Heterogeneity of impact on well-being

### 1.3.b Staggered DiD: static

Static effect

Table 4: Heterogeneity in static treatment effect estimates on well-being

|              | Strict only   |           | Multi only    |           | Both          |           |
|--------------|---------------|-----------|---------------|-----------|---------------|-----------|
| treat_on = 1 | 4.573         |           | -1.529        |           | 4.519         |           |
|              | (3.880)       |           | (2.442)       |           | (4.234)       |           |
| Num.Obs.     | 7367          |           | 9953          |           | 4246          |           |
| R2           | 0.003         |           | 0.001         |           | 0.006         |           |
| R2 Adj.      | 0.003         |           | 0.001         |           | 0.006         |           |
| AIC          | 69661.0       |           | 93978.9       |           | 41025.8       |           |
| BIC          | 69667.9       |           | 93986.2       |           | 41032.2       |           |
| RMSE         | 27.35         |           | 27.17         |           | 30.32         |           |
| Std.Errors   | Corrected     | Clustered | Corrected     | Clustered | Corrected     | Clustered |
|              | (cluster_uid) |           | (cluster_uid) |           | (cluster_uid) |           |

### 1.3.c Staggered DiD: dynamic event study

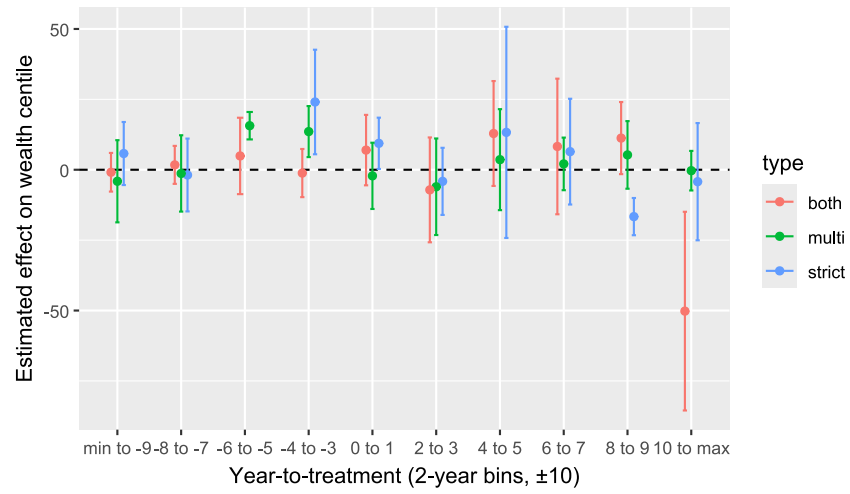


Figure 6: Heterogeneity of impacts on well-being: Event-study with 2-year bins